Experiment No. 12

Title

Tic Tac Toe Game Using Artificial Intelligence

Aim

To develop a Python program for implementing the Tic Tac Toe game using Artificial Intelligence.

Objective

The objective of this experiment is to understand the implementation of a Tic Tac Toe game and simulate player moves while checking for a winner or a draw.

Algorithm

Step 1: Initialize a 3×3 game board.
Step 2: Display the empty board.
Step 3: Allow players to enter their moves alternately.
Step 4: Update the board after each move.
Step 5: Check for a winning condition after every move.
Step 6: If no winner and the board is full, declare a draw.
Step 7: Stop.

Flowchart

```
        Start
          |
          ▼
   Initialize Board
          |
          ▼
   Display the Board
          |
          ▼
   Player Makes Move
          |
          ▼
    Update the Board
```

```
          |
          ▼
    Check for Winner
     |        |
    Yes       No
     |        |
     ▼        ▼
Display Winner   Board Full?
              |     |
             Yes    No
              |     |
              ▼     |
        Display Draw ——┘
              |
              ▼
            Stop
```

Python Program

```
board = [" " for _ in range(9)]


def display():
   print(board[0], "|", board[1], "|", board[2])
   print("--+---+--")
   print(board[3], "|", board[4], "|", board[5])
   print("--+---+--")
   print(board[6], "|", board[7], "|", board[8])


def check_winner(player):
   win = [
      (0,1,2),(3,4,5),(6,7,8),
      (0,3,6),(1,4,7),(2,5,8),
```

```
        (0,4,8),(2,4,6)
    ]
    for a,b,c in win:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

player = "X"

for _ in range(9):
    display()
    pos = int(input("Enter position (1-9): ")) - 1

    if board[pos] == " ":
        board[pos] = player

        if check_winner(player):
            display()
            print(player, "Wins!")
            break

        player = "O" if player == "X" else "X"
else:
    display()
    print("Match Draw")
```

Sample Output

```
  |   |
--+---+--
  |   |
--+---+--
```

| |

Enter position (1-9): 1

X | |

--+---+--

| |

--+---+--

| |

...

X Wins!

Result

The Tic Tac Toe game was successfully implemented, allowing two players to play and correctly identifying the winner or a draw.

Conclusion

Thus, the Tic Tac Toe game was successfully developed using Python. The program manages player turns, updates the game board, and accurately determines the game outcome based on the winning conditions.